PHS laboratory information note for mpox testing in Scotland

Version 8

Publication date: 23 Sept 2024

Contents

Version history 3

Scope 4

Background 4

Testing provision 5

Sample requirements 6

Transport and packaging 6

Important points about sample-labelling, packaging and request forms. 7

Laboratory safety 8

Notification of results 8

Appendix: Contact details 10

PHS Public Health Microbiology team 10

Edinburgh Specialist Virology Centre 10

West of Scotland Specialist Virology Centre 10

UKHSA 11

Version history

Version	Date	Summary of changes
V8.0	2309/24	Update to background section to reflect change in epidemiology Updates to reflect introduction of Clade testing in Scotland
V7.0	26/03/24	Updates to testing provision section to clarify different routes for HCID and non-HCID cases
V6.0	11/05/23	Change of name from monkeypox to mpox throughout when referring to the disease only Addition of scope section Updates to HCID categorisation within background section Updates to testing provision when referring suspected/confirmed HCID samples Updates to laboratory safety and notification of results sections, to clarify current pathways
V5.0	05/10/22	Updates to background section providing updated nomenclature and HCID information Updates to sample requirements, transport and packaging and notification of results sections to advise of process for testing, transport and reporting of suspected HCID case samples
V4.0	28/07/22	Addition of information around semen testing Changes to notification of results, including clarity on HPTs only disseminating positive results to relevant parties including PHS
V3.0	13/07/22	Changes to background section to link out to PHS Monkeypox page and to reflect HCID status derogation Changes to sample requirements section to include sample types Changes to transport and packaging section to reflect changes to sample transport requirements
V2.0	09/06/22	Changes following testing in Scotland introduced
V1.0	27/05/22	Original guidance published

Scope

This guidance is primarily aimed at clinical diagnostic laboratory staff. It should be used for all confirmed or suspected cases of mpox, irrespective of whether the case is being managed as a High Consequence Infectious Disease (HCID) or not.

Instructions for the collection of samples and appropriate PPE to be used during sample collection is out of scope for this document. IPC advice can be found in the **National Infection Prevention and Control Manual**.

Background

The **World Health Organization (WHO)** has recommended that 'monkeypox' should now be referred to as 'mpox'. The name change is to address concerns around stigmatising language associated with the disease.

As of 6 May 2022, public health agencies in the United Kingdom have responded to an outbreak of mpox. See **PHS mpox page** for current information and further generic guidance.

There are two known clades of MPXV: clade I and clade II. Transmission of mpox to humans can be due to zoonotic transmission or person-to-person spread. Infection with clade I MPVX has been reported to cause more severe mpox disease with a higher case fatality rate. The increased geographic spread of clade I mpox in Africa and emergence of clade Ib mpox has increased the need for mpox screening and typing in Scotland.

To differentiate between clade I and clade II, a single nucleotide polymorphism (SNP) based PCR approach has been adopted. This approach can also identify clade I (non 1b) from clade 1b without the need for sequencing, however samples can be subsequently sequenced if required (e.g. to support surveillance).

Testing provision

In Scotland, the Edinburgh Specialist Virology Centre (SVC) and the West of Scotland Specialist Virology Centre (WoSSVC) Glasgow, are the designated diagnostic laboratories for mpox testing.

- At present, WoSSVC Glasgow provides clade I (HCID) testing for all NHS Scotland boards.
- WoSSVC Glasgow provides Clade II (non-HCID) testing for NHS Ayrshire and Arran, NHS Dumfries and Galloway, NHS Forth Valley, NHS Grampian, NHS Greater Glasgow and Clyde, NHS Highland, NHS Lanarkshire, NHS Orkney, NHS Shetland and NHS Western Isles.
- SVC, Edinburgh provides clade II (non-HCID) testing for NHS Borders, NHS Fife, NHS Lothian and NHS Tayside. Clade I testing will be available at SVC Edinburgh from the 23rd of October 2024.

The requesting clinical team should discuss testing of those who meet the definition of a **HCID case** with the **Imported Fever Service (IFS)**, including testing of samples for other purposes prior to sending of sample(s). Where there is agreed clinical need, this testing may be performed locally at SVC/WOSSVC to prevent potential delay in diagnosis or management of other illness that may require urgent treatment.

SVC/WoSSVC should be informed via email by the requesting clinical team in advance of samples being submitted for testing. Contact information can be found in the Appendix.

Mpox testing is offered by WoSSVC/SVC seven days a week with one run performed daily. Mpox testing is not available as an on-call/out of hours service/assay.

- Samples which arrive before 9am will be tested on the same day, samples which arrive after 9am will be tested the following day.
- The turnaround time for clade testing from the receipt of sample is 1-2 days.

Sample requirements

Samples must be in molecular sampling solution (MSS), this has been **demonstrated to effectively inactivate mpox** and has been validated/verified for the testing systems in use. Other sample types require handling at Containment level 3. If not routinely held, Boards should consider having a small contingency stock of MSS for suspected mpox cases. If is it not possible to send samples in MSS, SVC/WOSSVC must be contacted for advice before sending.

The preferred sample is a swab of vesicular lesions. A throat swab is recommended for probable cases who have developed systemic symptoms but do not have a rash or lesions for sampling. For individuals presenting with proctitis alone, a rectal swab can be taken.

Semen testing may be indicated in a very limited number of situations – see **UKHSA mpox: semen testing for viral DNA guidance** for further details. This is not a validated sample type for local assays and testing is not performed in Scotland. Please see Appendix for UKHSA contact details for referral of specimens. Results will be shared with the requesting clinician/laboratory directly from UKHSA.

ALL samples should be clearly labelled and handled with appropriate precautions.

Cut off for receipt of samples is 9am and all efforts should be made for samples to reach SVC/WoSSVC in advance of this time. SVC/WoSSVC should be informed via email in advance of samples being submitted for mpox testing, so that the appropriate precautions can be taken to minimise risk to laboratory workers.

Samples from confirmed cases should not be sent for follow up testing unless there is a clearly defined clinical need.

Transport and packaging

Where the case is a suspected HCID, this should also be added to the request form.

- For samples to be sent to the SVC for mpox testing, requests should be sent through NPex or if no access to this, then the following **request form** should be completed (select RIE and Virology to be taken to list of tests with mpox form).

- For samples sent to the WoSSVC, use the following generic **request form**.

Important points about sample-labelling, packaging and request forms.

- Local laboratories are to ensure initial contact is made with local infectious diseases colleagues to determine if any cases meet the definition of a HCID. For potential HCID cases, the IFS should always be contacted prior to requesting testing.

- Email the SVC/WoSSVC to inform of the reason for testing and contact details for result return. See contact details in Appendix. This process should be used for both in hours and out of hours requests (which will be picked up in the following morning including at weekends).

- Ensure that all stoppers and lids are firmly attached. Samples that have leaked in transit cannot be processed if there is insufficient material left.

- ALL samples should be clearly labelled identifying a suspected mpox case. This includes samples to be sent for mpox testing AND samples to be sent for other investigations.

- Each sample must be labelled with ID, date of birth and type of sample.

- Paperwork (request forms) should NOT be placed in the primary container for Cat B transport.

Samples should be packaged and transported in accordance with Category B transportation regulations. UN 3373 packaging must be used for sample transport.

Where possible, suspected clade I samples should be transported by a direct Category B courier.
For suspected clade II samples, transport may be through DX or by Category B courier.

Laboratory safety

Mpox is designated a human pathogen hazard group 3 by the **Advisory Committee on Dangerous Pathogens.**

All sample handling should only be undertaken following local risk assessment. Further details can be found in **UKHSA guidance for mpox diagnostic testing - Information for laboratories.**

Waste associated with Mpox testing should be managed as "Category B" (infectious/healthcare waste).

Notification of results

SVC/WOSSVC will advise of positive and negative results from samples referred to them for mpox testing by telephone or email to the requesting clinician or person indicated on the email. If a PCR positive clade I case is identified, UKHSA should be informed by the WoSSVC/SVC.

It is the responsibility of the clinician caring for the patient to notify the local HPT whether samples are positive or negative. Notification should be made promptly on receipt of the result and should include full patient details. This will enable appropriate public health actions to be taken for positive results, or to be stood down for negative results, without delay.

HPT's can manually add the case as suspected to HPZone in anticipation of a test result being sent. Test results will be shared with ECOSS automatically from the SVC/WoSSVC. Positive results will be shared with HPZone.

The local HPT should inform Public Health Scotland and other relevant parties within their health board of any positive results, including the local NHS Scotland diagnostic microbiology and virology laboratory who may be testing other samples.

HPTs should aim to complete an interview and enhanced surveillance form within 48 hours of notification of the case. Enhanced surveillance forms can be downloaded from **SHPIR**. Completed forms should be returned to the BBVSTI team at PHS: **phs.bbvsti@phs.scot**

Any result enquiries should be sent to (as appropriate):

Edinburgh: virologyadvice@nhs.scot

Glasgow: west.ssvc2@nhs.scot

RIPL: ripl@ukhsa.gov.uk (do not include patient identifiable information)

Appendix: Contact details

PHS Public Health Microbiology team

Email: **PHS.PHM_Team@phs.scot**

Edinburgh Specialist Virology Centre

Specialist Virology Centre
Royal Infirmary of Edinburgh
51 Little France Crescent
Edinburgh
EH16 4SA
DX address DX 6231202, Edinburgh 96EH

Email: **virologyadvice@nhs.scot**
Phone (9am - 5pm, Monday to Friday): 0131 536 3373 (option 2)

West of Scotland Specialist Virology Centre

Virology Reception (Level 5)
New Lister Building
Glasgow Royal Infirmary
10-16 Alexandra Parade
Glasgow
G31 2ER
DX address DX 6491304, Glasgow 94G

Email: **west.ssvc2@nhs.scot**
Phone (9am - 5pm Monday to Friday): 0141 201 8721

UKHSA

Rare and Imported Pathogens Laboratory
UKHSA
Manor Farm Road
Porton Down
Salisbury SP4 0JG

DX 6930400, Salisbury 92 SP

Email: **ripl@ukhsa.gov.uk** (do not include patient identifiable information)

Hospital doctors can contact the Imported Fever Service (IFS) after discussion with the local microbiology, virology or infectious disease consultant.

Phone: 0844 778 8990